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SUPERSEDING
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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M20A1 ASSEMBLY

1. SCOPE

1.1 This is the applicable detail specification for Thruster, Cartridge Actuated, M20A1 Assembly, loaded with Cartridge, Thruster, M127 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for
- MIL-I-45208 - Inspection System Requirements
- MIL-P-19834 - Plates, Identification, Metal Foil, Adhesive Backed

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MS-28741 - Hose Assembly-Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command
- F10533856 - Thruster, Cartridge Actuated M20A1, Assembly
- C8797741 - Container, Ammunition, Fiber M347 for Thruster, Cartridge Actuated, M20A1 or M5A3
- C8797742 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M20A1, or M5A3 in Fiber Container M347

FSC 1375

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DRAWINGS (Cont.)

IEL-10533856 - Inspection Equipment List for Thruster, Cartridge Actuated, M20A1

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Material, parts and assemblies shall comply with all the requirements specified on the applicable drawings and specifications.

3.2 Torque requirements.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

	A	B	C
Components	Run On (Inch pounds) (Minimum)	Seating or Bottoming (Including A) (Inch pounds)	Breakaway (Inch pounds) (Minimum)
Head and Retainer Guide	50	300 ± 20	200
Retainer Guide and Sleeve after insertion of cartridge	--	300 ± 20	200
Head and Body	50	600 ± 20	400

3.3 Assembly.

3.3.1 Trunnion rotation.- The trunnion shall be free to rotate 360 degrees.

3.3.2 Residual magnetism.- The thruster shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 The thruster shall not fail to fire.

3.4.2 The thruster shall function without mechanical failure (see 6.1).

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3.4.3 Locked shut.- The thruster fired in a locked shut condition (see 6.2) shall show no mechanical failure (see 6.1).

3.4.4 Load.- The thruster fired under load (see 6.3) shall perform as follows:

3.4.4.1 Thrust.- With the thruster connected to a 50 pound weight, the thrust shall move a constant load of 3000 pounds, horizontally, throughout its entire stroke of 5 inches, minimum.

3.4.4.2 Operating time.- The time from initial thrust rise to completion of the piston stroke shall be 30 milliseconds, minimum, and 70 milliseconds, maximum.

3.4.5 Ignition delay.- The ignition delay time shall not exceed 65 milliseconds (see 6.4).

3.4.6 Temperature.- The thruster shall comply with all ballistic requirements throughout the temperature range of minus 65 to plus 200 degrees Fahrenheit (F).

3.4.7 Initiation.- The thruster shall be initiated with an M3A1 Initiator connected to the thruster with 15 feet of aircraft hose, size 4, Standard MS-28741.

3.5 Workmanship.- The requirements for workmanship shall be as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components shall be free from cracks, splits, perforations or any similar defect.

3.5.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

3.5.4 Label, edge adhesion.- The labels shall withstand two or more traverses of a knife at any point without damage. There shall be no indication or poor adhesion; such as nicking, tearing or peeling of the labels in whole or in part from the thruster.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial

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laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with Specifications MIL-A-2550 and MIL-I-45208, except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with Standard MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Thruster, M127 Dwg D10520589, covering a detail of Dwg F10533856.

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Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual

4.2.2.2 Retainer, Guide Sub-Assembly consisting of Guide Retainer and Pellet Assembly 10521616, Shear Pin 10521618, Firing Pin 8593864 and O-Ring 501220 covering a detail of Dwg F10533856.

Defect	Method of Inspection
CRITICAL	
1. Striker hole obstructed	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Head Sub-Assembly consisting of Head 10521455, Guide Retainer and Pellet Assembly 10521616, Sleeve 10521617, Shear Pin 10521618, Firing Pin 8593864, Cartridge 10520589 and O-Rings 501220, 501228, covering a detail of Drawing F10533856.

CRITICAL	
None defined	
MAJOR	
101. Cartridge loose or missing	Manual/Visual
102. Thread damaged or not full	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

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Defect	Method of Inspection
MINOR	
201. Excess sealant not removed	Visual

4.2.2.4 Assembly, Drawing F10533856 less Shipping Cap 8593296 and Plug 8593299.

CRITICAL	
1. Shear pin not engaged in firing pin	Gage
2. Foreign matter in gas port	Visual
MAJOR	
101. Total length	Gage
102. Thread damaged	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F10533856.

CRITICAL	
1. Mixed cartridges	Visual
2. Mixed thrusters	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
None defined	
MINOR	
201. Trunnion not free to rotate 360 degrees	Manual
202. Total length	Measure
203. Location of labels incorrect	Visual
204. Shipping cap or plug missing	Visual

4.2.2.6 Fiber Container, Drawing C8797741.

CRITICAL	
None defined	
MAJOR	
None defined	

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Defect	Method of Inspection
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Box, Packing, Drawing C8797742.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3 except as provided in Specification MIL-I-45208.

4.2.3.1 Curing time.- Each retainer guide and sleeve sub-assembly shall be subjected to the curing time as specified on the drawing. Failure of any unit to comply with the requirement of the drawing shall be cause for rejection of the sub-assembly.

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4.2.3.2 Breakaway torque.

4.2.3.2.1 Retainer guide and sleeve.- Each retainer guide and sleeve sub-assembly shall be subjected to a breakaway torque application. Failure of any unit to comply with the requirements of 3.2 and Table I shall be cause for rejection of the sub-assembly.

4.2.3.2.2 Head and retainer guide.- Each head and retainer guide sub-assembly shall be subjected to a breakaway torque application. Failure of any unit to comply with the requirements of 3.2 and Table I shall be cause for rejection of the sub-assembly.

4.2.3.2.3 Head and body.- Each head and body assembly shall be subjected to a breakaway torque application. Failure of any unit to comply with the requirements of 3.2 and Table I shall be cause for rejection of the assembly.

4.2.3.3 Residual magnetism.- Each thruster shall be tested for excessive residual magnetism. Failure of any thruster to comply with the requirements of 3.3.2 shall be cause for rejection of the thruster.

4.2.3.4 Label, edge adhesion.- Ten assemblies from each lot shall be tested for adhesion of the label edges. Failure of any unit to comply with the requirements of 3.5.4 shall be cause for rejection of the lot of thrusters.

4.2.3.5 Functional tests.- The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics.- Sixteen units shall be tested to determine compliance with the requirements specified in 3.4.1 through 3.4.7. The unit shall be temperature conditioned as provided in Table III and 4.3.5.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

Table III

Sample Size	Method	Temperature
7	Load	-65° to -70° F
7	Load	200° to 205° F
1	Locked Shut	-65° to -70° F
1	Locked Shut	200° to 205° F

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10533856, except as provided in Specification MIL-I-45208.

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4.3 Test methods and procedures.

4.3.1 Curing time.- After insertion of cartridge the thread sealing compound shall be applied to the threads on the sleeve in accordance with the applicable drawing.

4.3.2 Breakaway torque.- A torque wrench set at the torque requirements as specified in 3.2, Table I shall be used in conjunction with a fixture to determine the adequacy of the joints.

4.3.3 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading as shown in Figure 1. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

4.3.4 Label, edge adhesion.- The label edge adhesion test shall be in accordance with Specification MIL-P-19834.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The sample shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded the sample units shall be reconditioned at the original temperatures.

4.3.5.1.1 Temperature conditioning.- A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature. The total conditioning time including reconditioning at plus 200 degrees F. shall not exceed 18 hours.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.- Packing and marking shall be in accordance with Drawings C8797741& C8797742.

5.2 Data cards.- Data card information shall be as specified in Specification MIL-A-2550.

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6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Locked shut.- Locked shut is defined as a condition in which the piston is restrained from moving out of the thruster proper upon firing.

6.3 Load.- The expected performance of the thrusters under conditions simulating those existing in aircraft is obtained when thrusters satisfactorily meet the conditions of the load firing.

6.4 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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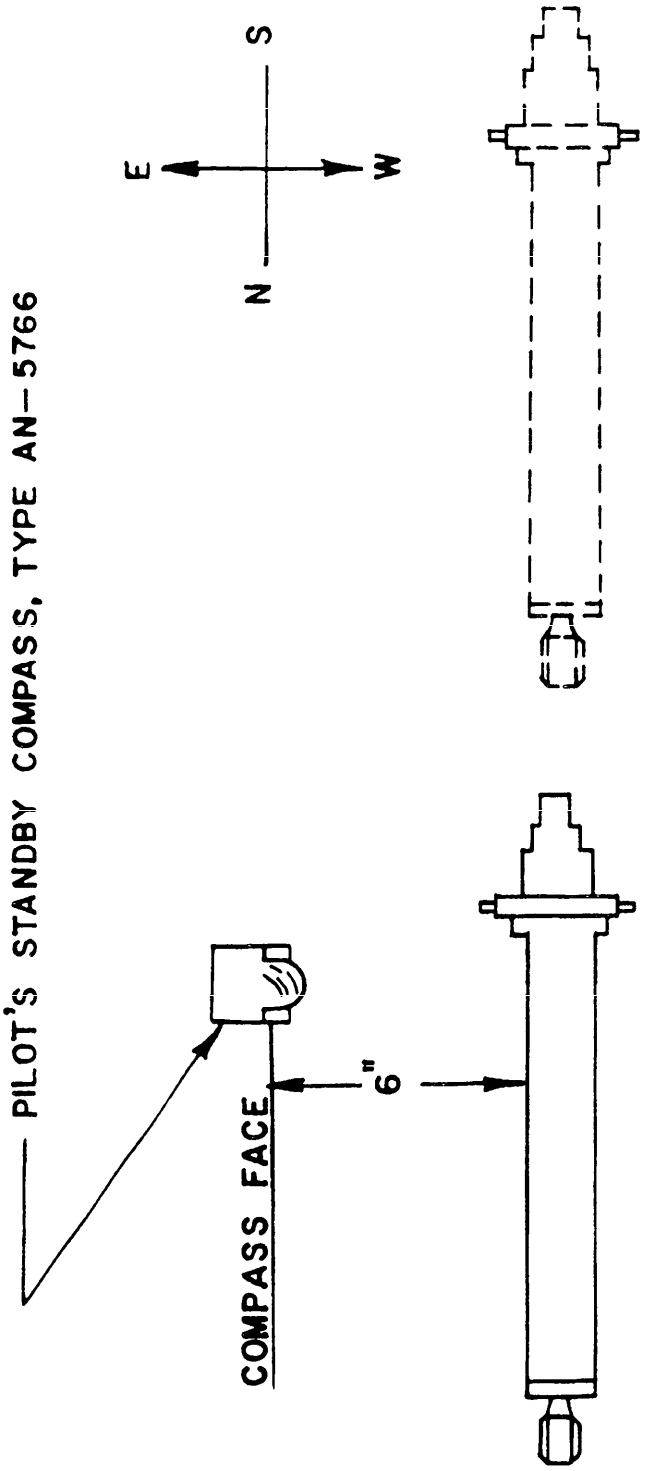
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NOTE:
THRUSTER AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. LAYOUT FOR DETERMINATION FOR RESIDUAL MAGNETISM IN THRUSTER, CARTRIDGE ACTUATED, M20A1.

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